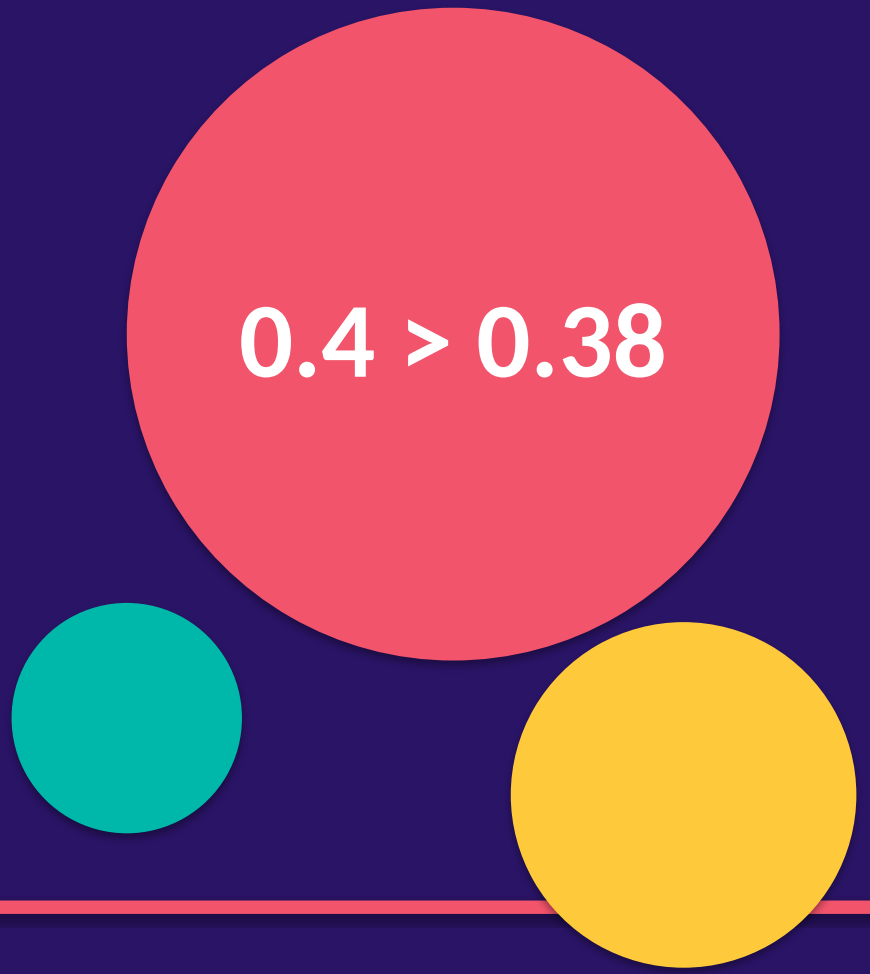
Understanding and ordering decimals

Year 10 · Edexcel GCSE Foundation · 60 minutes · Mastery lesson



LO1: Know the value of each digit in a decimal
LO2: Compare two decimals
LO3: Put decimals in order

Lesson for Sienna · 60 minutes · OnMaths · Mr Ali

Have ready: mini whiteboard, pen, calculator OFF

Do Now

8 minutes

On your own. Show your working. Then we check together.

Last lesson: $-3 - 5$

Last lesson: -4×-6

Number skill: order $-2,\ 5,\ -7$

Talk Task

5 minutes

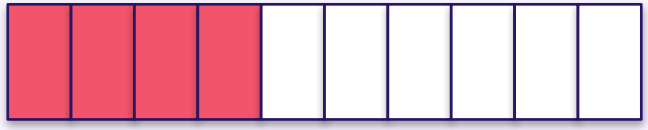
Always, Sometimes or Never?
“A decimal with more digits is bigger.”

Say: I think it is _____ because _____ . For example, _____ .

Key facts: place value

Key facts

Ones		Tenths	Hundredths	Thousandths
3	.	4	7	2



4 tenths = 0.4

3.472 = 3 ones, 4 tenths, 7 hundredths, 2 thousandths

Tenth	0.1 – 1 whole split into 10
Hundredth	0.01 – 1 whole split into 100
Thousandth	0.001 – 1 whole split into 1000
Decimal point	separates the whole number from the parts

LO1 I do: the value of a digit

I do

What is the value of the 7 in 3.472?

Ones		Tenths	Hundredths	Thousandths
3	.	4	7	2

LO1 We do

We do

What is the value of this digit? Say the column name.

the 6 in 2.36

the 9 in 14.9

the 4 in 0.004

4 tenths + 7 hundredths

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO1 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

Value of the 8 in 6.82

Value of the 3 in 0.163

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO1 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

2 ones + 5 hundredths

6 tenths + 3 thousandths

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO1 You do

B · S · G · P

BRONZE

What is the value of the 6 in 2.65?

SILVER

What is the value of the 9 in 4.209?

GOLD

Write the decimal with 3 tenths,
0 hundredths and 8 thousandths.

PLATINUM

Use the digits 4, 7 and 1 once each.
Make the biggest decimal less than 1.

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO2 I do: compare two decimals

I do

Which is bigger: 0.6 or 0.58?

Ones		Tenths	Hundredths	Thousandths
0	.	6		

Ones		Tenths	Hundredths	Thousandths
0	.	5	8	

LO2 We do

We do

< , > or = ? You say it first, then click.

3.4 ☐ 3.39

0.07 ☐ 0.1

2.5 ☐ 2.50

0.909 ☐ 0.91

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO2 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

Which is bigger: 0.3 or 0.29?

Which is bigger: 1.08 or 1.8?

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO2 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

$0.45 \square 0.405$

Which is more: £2.5 or £2.05?

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO2 You do

B · S · G · P

BRONZE

Which is bigger: 0.7 or 0.65?

SILVER

Write < or >:

4.06 ☐ 4.6 0.8 ☐ 0.78

GOLD

Amy ran 100 m in 13.45 s. Ben took 13.5 s.
Who was faster? Explain.

PLATINUM

Sam says 0.25 > 0.3 because 25 > 3.
Explain his mistake.

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO3 I do: order decimals

I do

Put in order, smallest first: 0.4, 0.38, 0.405, 0.04

LO3 We do

We do

Fill the gaps with zeros, then order smallest first.

2.1, 2.01, 2.11, 2.101

1.5 kg, 1.45 kg, 1.054 kg

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO3 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

Order: 0.6, 0.06, 0.66

Order: 3.2, 3.02, 3.22

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO3 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

Largest: 0.71, 0.7, 0.077

Smallest: 5.5, 5.05, 5.055

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

LO3 You do

B · S · G · P

BRONZE

Order, smallest first:
0.3, 0.13, 0.31

SILVER

Order, smallest first:
4.5, 4.05, 4.55, 4.505

GOLD

Heights: Ali 1.6 m, Bea 1.56 m,
Cal 1.605 m, Dee 1.65 m.
Order them, tallest first.

PLATINUM

Order, smallest first:
−0.5, 0.05, −0.55, 0.5

Help: line up the decimal points

Ones		Tenths	Hundredths	Thousandths
	.			

Fill gaps with 0

Exam questions

Edexcel style

1. **Write these numbers in order of size.**
   Start with the smallest. 0.62 0.6 0.059 0.612

(1 mark)

2. **Write down the value of the 4 in 18.346**

(1 mark)

3. **Write down a decimal between 0.3 and 0.4**

(1 mark)

Exit ticket

5 minutes

On your own. Also on your OnMaths page, where it is marked for you.

1. **Which is bigger: 0.8 or 0.75?**

2. **Which is the smallest: 0.45, 0.405, 0.5, 0.054?**

3. **Write the decimal with 6 tenths and 3 thousandths.**

Finished early? Extension

Extension

Which decimal is exactly halfway between 0.4 and 0.5?

Use 2, 5, 0 and a decimal point once each.
Make the smallest number bigger than 1.

Order: 0.7 m, 65 cm, 0.072 m

Homework

Due next Friday

4 questions on understanding and ordering decimals (6 marks).

Do it online on your OnMaths page – it marks itself.
Show your working on paper.

Stuck? Draw a place value grid and fill gaps with zeros.

Today I learned: value of a digit ✓ compare decimals ✓ order decimals ✓